

Measuring the Tor network

hiro arma

August 2024

The Tor Project





By Hamed Saber

Today's talk

- (1) What is Tor, a brief introduction
- (2) The Tor relay community—understanding and growing the Tor network
- (3) The Tor network in metrics
- (4) How to get involved

What do we mean by *Tor*?



Ecosystem of
privacy preserving
technologies



Non-profit
organization



Global community
of volunteers and
supporters

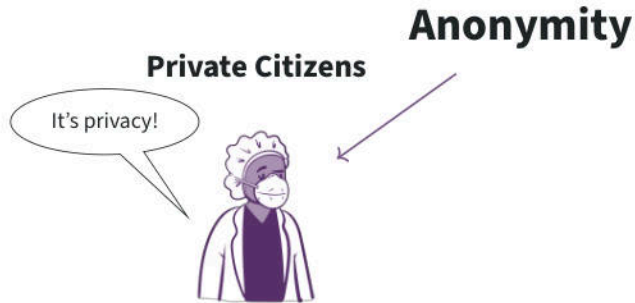
Communications Metadata



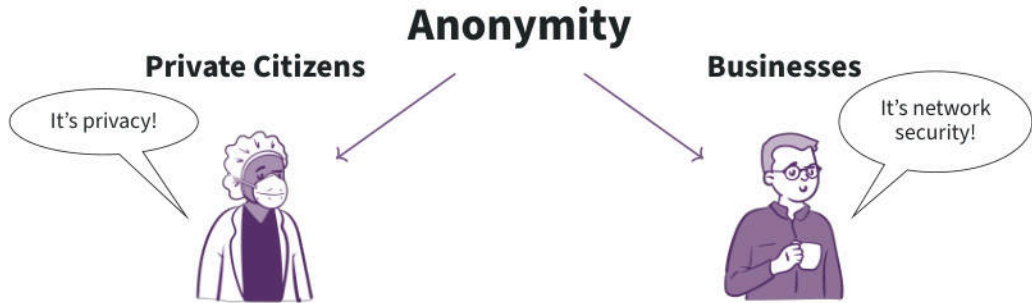
“We Kill People Based on Metadata.”

—Michael Hayden, former director, NSA

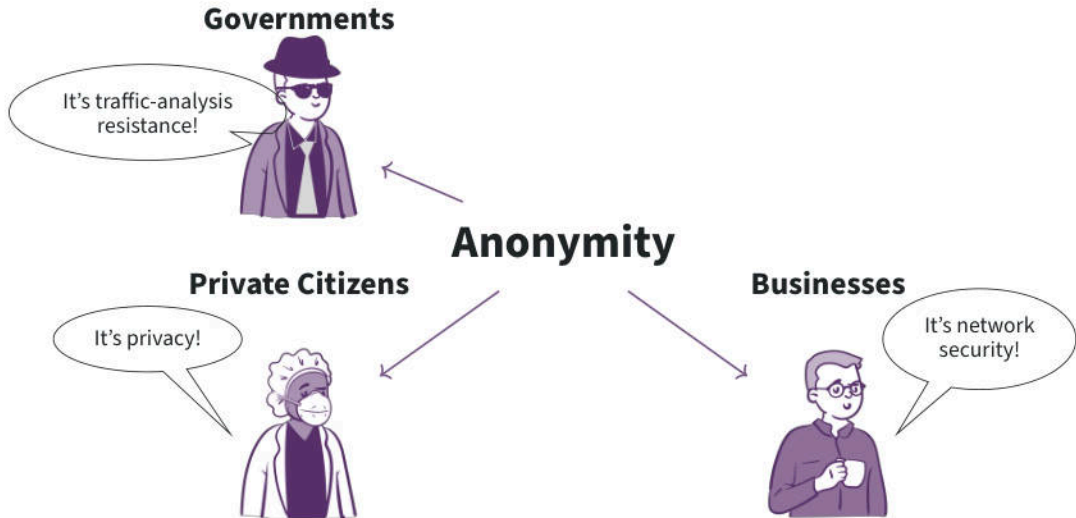
Anonymity is different for each use case



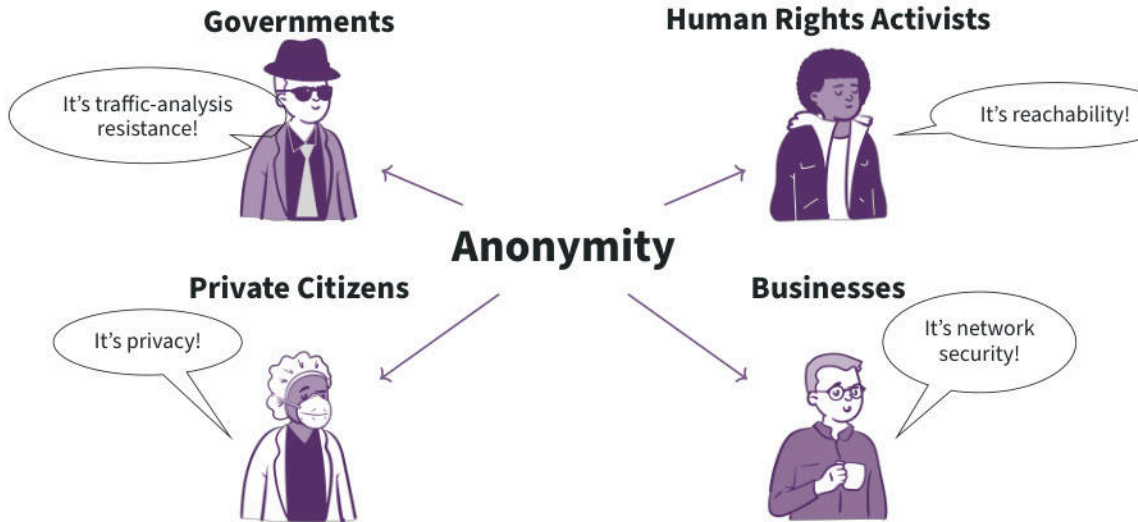
Anonymity is different for each use case



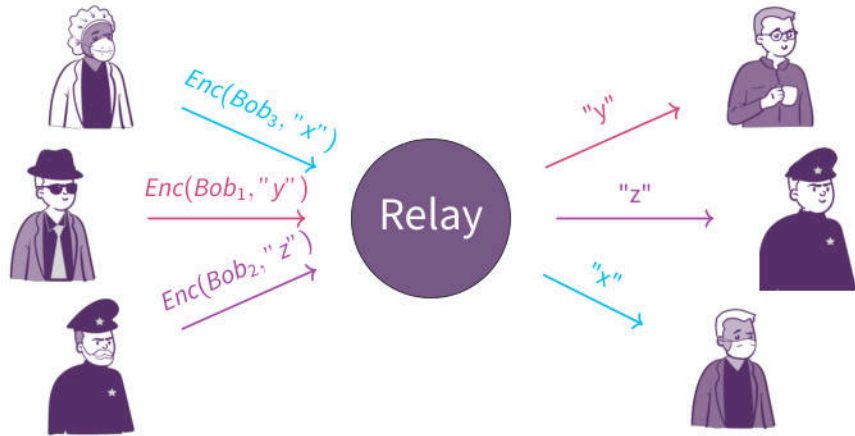
Anonymity is different for each use case



Anonymity is different for each use case

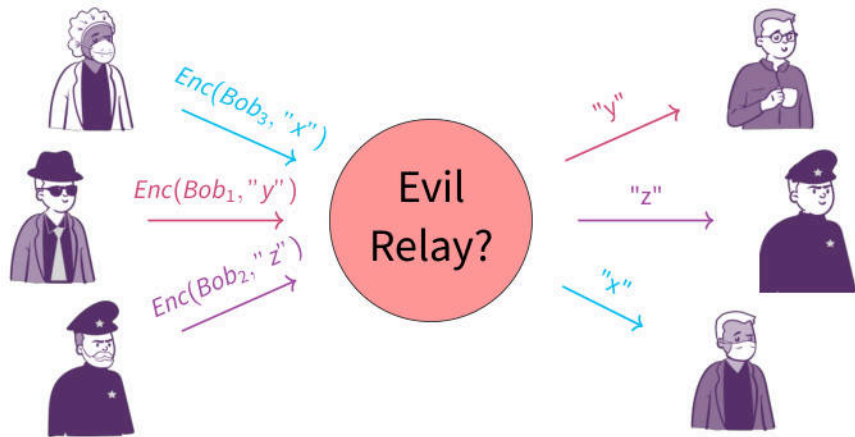


A Simple Design

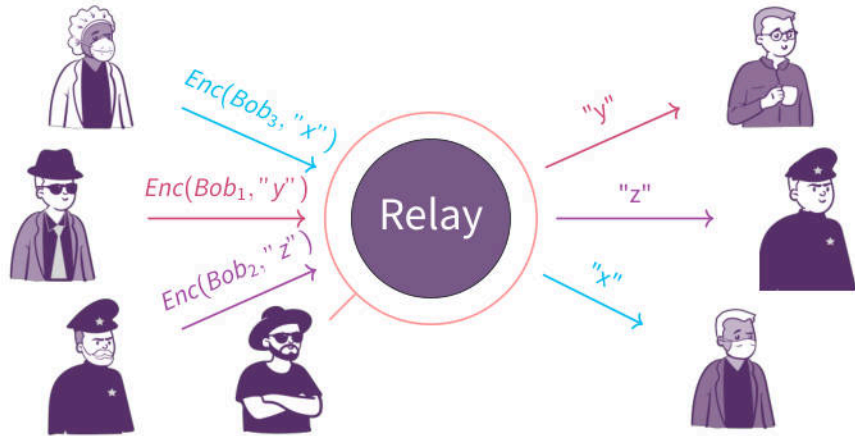


Equivalent to most commercial proxy/VPN providers

A Simple Design



A Simple Design



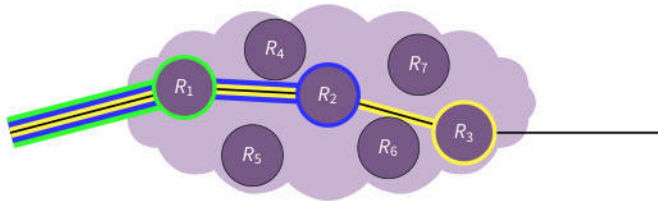
Timing analysis lets an observer match up connections

The Tor Design

Alice



Anonymity Network



Bob



Multiple relays so no single relay can link Alice to Bob

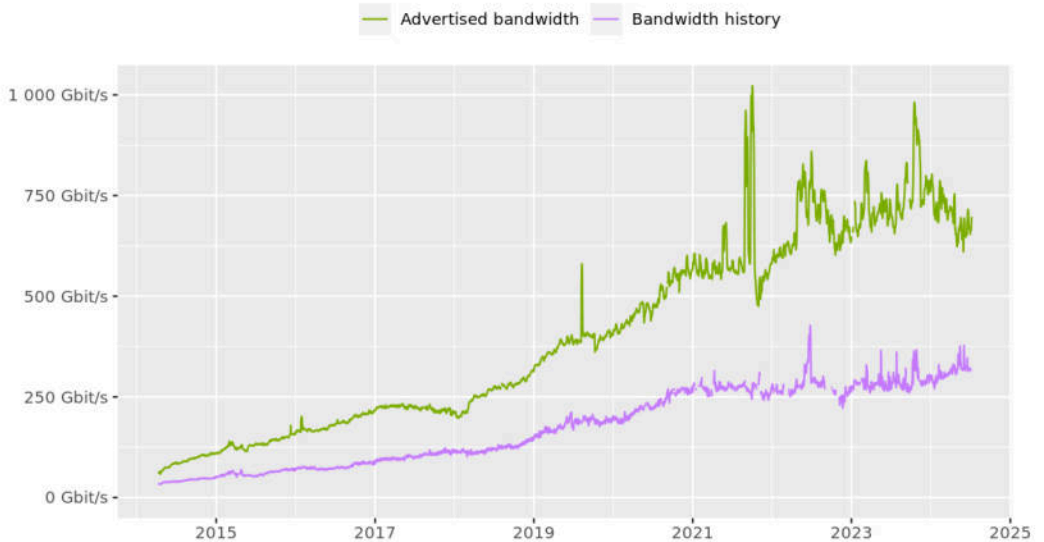
Transparency for Tor is key

- Open source / free software
- Public design documents and specifications
- Publicly identified developers

Transparency for Tor is key

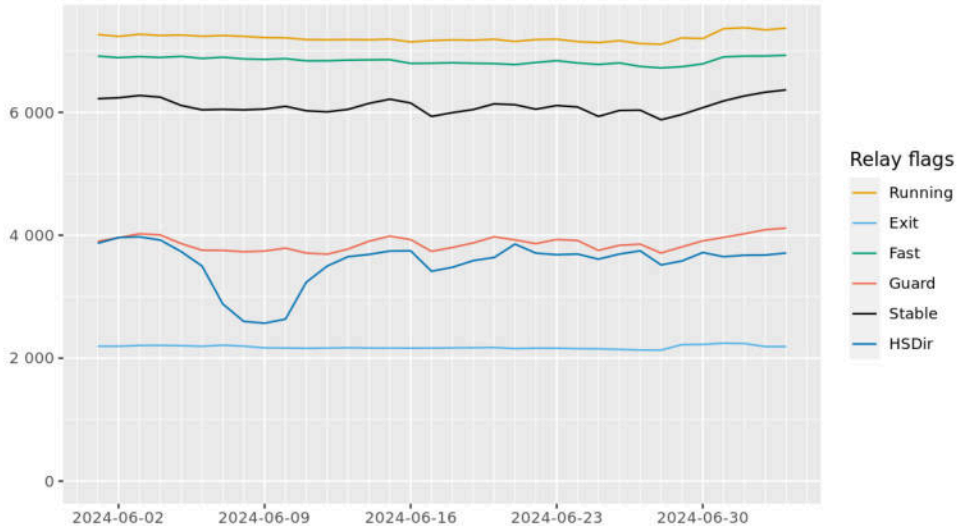
- Open source / free software
- Public design documents and specifications
- Publicly identified developers
- Not a contradiction: privacy is about choice!

Total relay bandwidth



Understanding the Tor network

Number of relays with relay flags assigned



Understanding the Tor network

The Tor network is driven by a diverse **community**

Diversity within the Tor network is a critical component encompassing various desired properties

Types of hardware, software, configurations, location, users

Understanding the Tor network

Fostering a healthy community

- **direct** communication
- online and in person **meetups**
- **transparent policies**
 - community.torproject.org/policies/relays

Relay landscape, past/present/future

For the first decade, many relay operators had met me at a conference

Then the set of reasons to offer well-funded harmful relays expanded

- (1) Need to improve applications, e.g. https by default
- (2) One of the few tools we have for safety in this asymmetric situation is our social graph

Relay operator associations

Many Tor exit relays are run by nonprofits that people created to operate relays in their country

Successful in Switzerland, Netherlands, US, Germany, Sweden, France, Canada, Austria, Denmark, more

Even better is that they're part of our community: they're connected to hacker spaces and we met them at conferences like this one



I WANT TO RUN A RELAY

YOU'LL NEED TO DO THE FOLLOWING

1

CHOOSE A RELAY

2

FIND ALLIES

3

GET A PRIZE

1

CHOOSE YOUR RELAY



Relay incentives

Gamification / badges / leader boards

Meetups, attentive operator support

Structural incentives, like better speed for relay operators, or everybody-a-relay

...Relay operator payments?

**EL PROBLEMA
ES EL CAPITALISMO**

Yo

**una conversación
casual**

The Tor network in metrics

In May 2020, we removed a group of Tor exit relays that were interfering with exit traffic by intercepting connections to a small number of cryptocurrency exchange websites

[*] blog.torproject.org/bad-exit-relays-may-june-2020

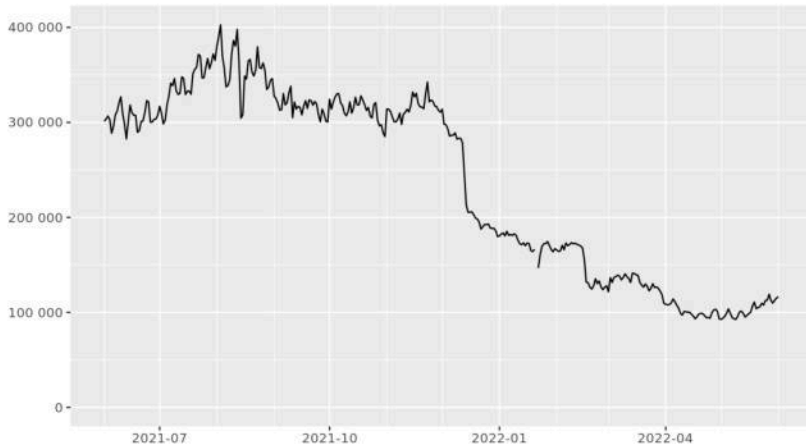
The Tor network in metrics

The Network Health team is dedicated to monitoring our network, encompassing a joint effort between various teams at Tor

The foundation of this work is the metrics that provide us with the status of the network

Start of the Ukraine invasion 2021-2022

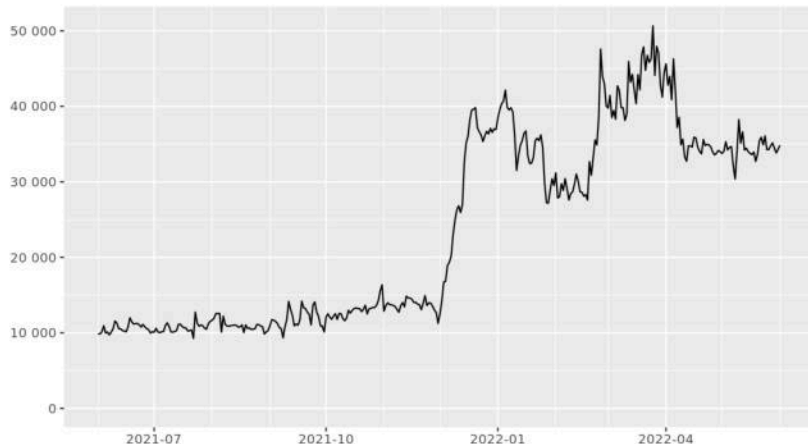
Directly connecting users from Russia



The Tor Project - <https://metrics.torproject.org/>

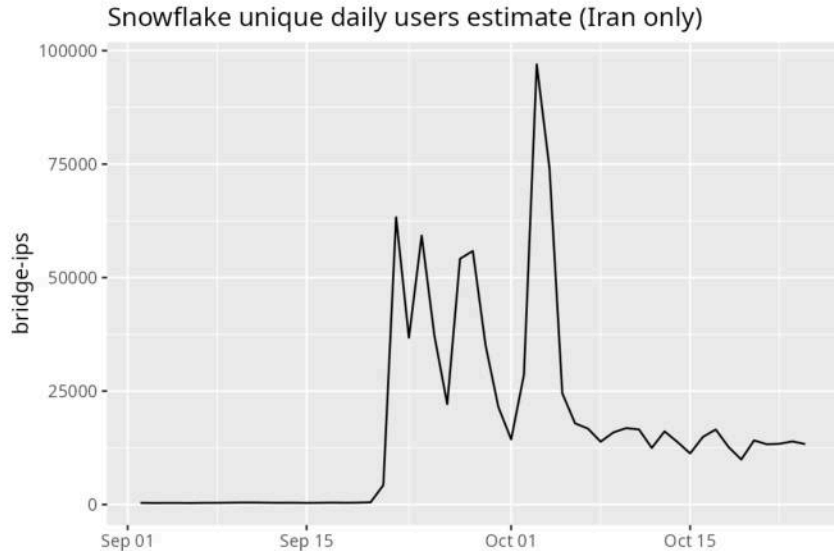
Start of the Ukraine invasion 2021-2022

Bridge users from Russia



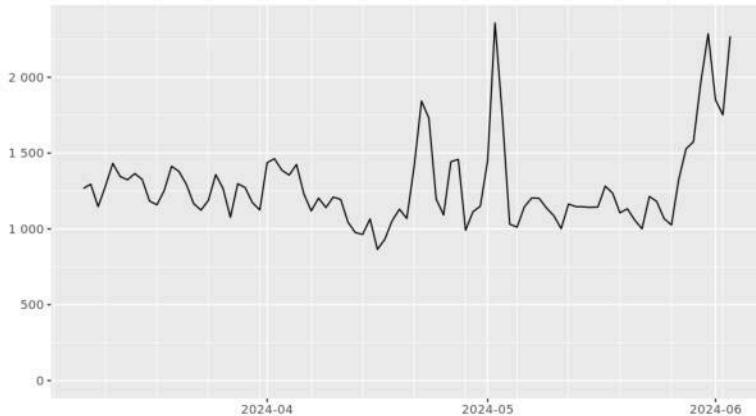
The Tor Project - <https://metrics.torproject.org/>

Mahsa Amini protests in Iran 2022



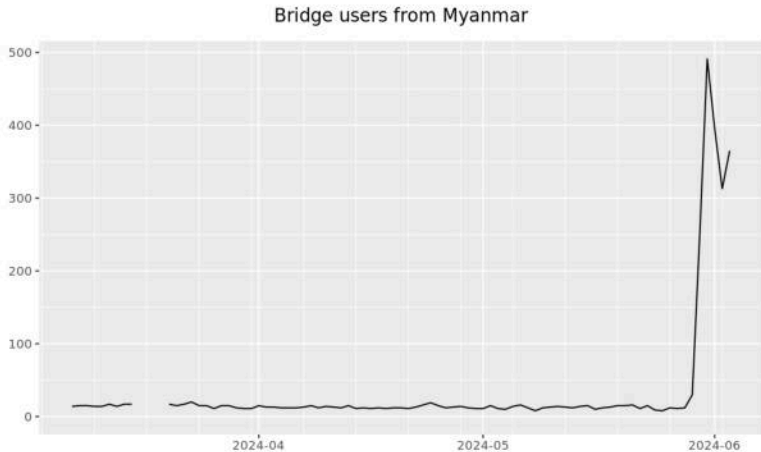
Internet censorship in Myanmar May 2024

Directly connecting users from Myanmar



The Tor Project - <https://metrics.torproject.org/>

Internet censorship in Myanmar May 2024



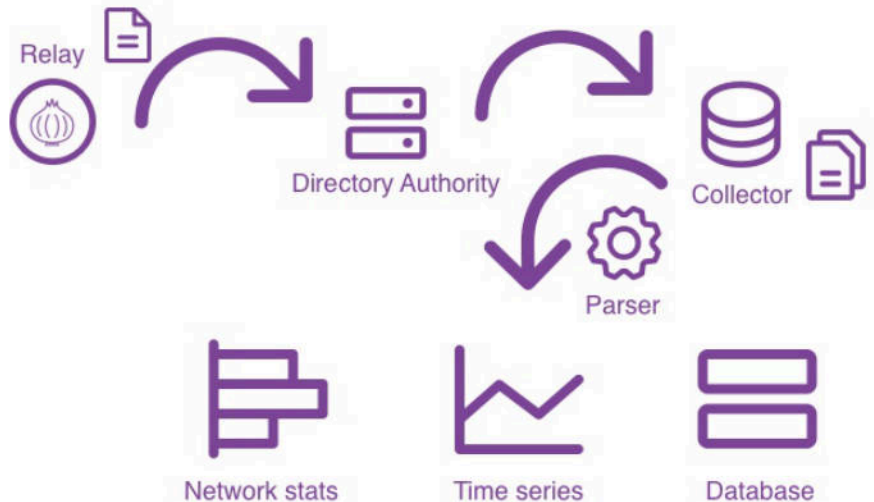
The Tor Project - <https://metrics.torproject.org/>

The Tor network in metrics

Do not collect data that cannot be public

- Do not collect in client data
 - Collect aggregated data from relays
- Do keep historical data (collector.torproject.org)
 - Network wide statistics
 - Status of individual nodes (bridges, relays)
- Accessible data and algorithm transparency

The Tor network in metrics



The Tor network in metrics

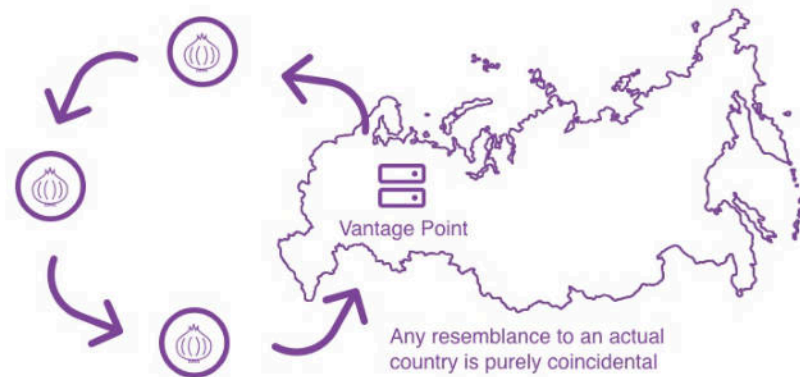
- Traffic/usage patterns
- Bandwidth measurements
- Performance measurements

Bandwidth probing



Bandwidth is measured from the bandwidth authority by creating a circuit to the relay and downloading a file from a web server

Performance measurements



Runs a Tor client in a censored region, spins up a web server and downloads 5MB files over the Tor network

Metrics data is open and accessible, but there is a learning curve

Our new API lowers the barrier for researchers to start working with Tor network data

Few endpoints (more coming up soon):

- metrics-api.torproject.org/details
- metrics-api.torproject.org/summary

metrics-api.torproject.org

```
{  
  "nickname": "seele",  
  "fingerprint": "000A10D43011EA4928A35F610405F92B4433B4DC",  
  "last_seen": "2024-08-02 13:00:00",  
  "running": true,  
  "flags": ["Fast", "HSDir", "Running", "Stable", "V2Dir", "Valid"],  
  "country_name": "United States of America",  
  "as": "AS7018",  
  "as_name": "AT&T Services, Inc.",  
  "last_restarted": "2024-07-18 11:59:51",  
  "bandwidth_rate": 1048576,  
  "bandwidth_burst": 2097152,  
  "observed_bandwidth": 1062365,  
  ...  
}
```

[gitlab.torproject.org/tpo/network-health/
metrics/networkstatusapi/-/tree/main/docs](https://gitlab.torproject.org/tpo/network-health/metrics/networkstatusapi/-/tree/main/docs)

Get involved: run a relay!

More relays means better performance for users

And more good relays dilute any bad relays

Location matters, among many other diversity factors

Learn about exit policies, and learn about IP blocklists

Get involved: run a relay!

<https://community.torproject.org/>

Join the community, come to the relay operator meetups

If you can't run a relay, run a bridge or a Snowflake!

Get Involved: Help with Network Health!

- Network data analysis
- **Tor Weather** – A tool for relay operators
- Get involved with the redesign of our metrics data pipeline
 - **Network Status API** – Access real-time network data.
 - **descriptorParser** – Parse network documents in Java.
 - **TagTor** – Visualize network data and metrics.
 - **tor_fusion** – Parse network documents in Rust.
- gitlab.torproject.org/tpo/network-health/team/-/wikis/home#how-to-get-involved